Tahtabir Rahman

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Professional Summary

Data Engineer with **4+ years** of experience in data engineering and analytics, building reliable **ETL/ELT pipelines** that turn raw data from multiple sources into trusted, analytics-ready datasets. Currently processing **50–100 GB** of data daily across **5 source types** using **Python, PySpark, SQL, Azure Data Factory, and Azure Databricks**, with working experience on **AWS**. Strong in data quality, reconciliation, and pipeline reliability, and proven at automating manual data work (**30%** reduction in manual data handling). Partners with engineering and business teams to deliver dependable data that supports reporting, BI, and decision-making.

Technical Skills

Languages: Python (Pandas, NumPy, PySpark), SQL
Data Pipelines & ETL/ELT: Azure Data Factory, Azure Databricks, Apache Spark, Batch Processing, Pipeline Automation
Cloud Platforms: Microsoft Azure, AWS (S3)
Data Quality: Validation, Cleansing, Reconciliation, Exception Handling, Profiling
Data Modeling & SQL: Data Modeling, Joins, CTEs, Window Functions, Query Optimization
DevOps & Version Control: Git, GitHub, CI/CD
BI & Reporting: Power BI, Tableau, Advanced Excel, KPI Dashboards
AI Tools: ChatGPT, Claude, Microsoft Copilot, Google Gemini

Experience

Data Engineer — India
S-Cube Solutions Private Limited — *Nov 2025 – Present*
Project: Enterprise Data Integration & ETL/ELT Pipeline Development
Tech Stack: Python, SQL, PySpark, Azure Data Factory, Azure Databricks, AWS, Git, CI/CD

- **Delivered analytics-ready data at scale:** Designed and maintained daily ETL/ELT pipelines (Python, PySpark, SQL) processing **50–100 GB** across **5 source types**: relational databases, CSV, JSON, REST APIs, and cloud storage.
- **Automated end-to-end data delivery:** Built batch ingestion and transformation workflows in Azure Data Factory and Databricks, coordinating dependencies between stages for consistent downstream reporting.
- **Improved processing efficiency:** Optimized SQL queries and PySpark transformations by reviewing execution details and removing redundant processing.
- **Protected data trust:** Developed data-quality and reconciliation checks with exception handling to catch missing, duplicate, and inconsistent records before they reach reports.
- **Strengthened reliability:** Monitored pipeline runs, diagnosed recurring failures through root-cause analysis, and implemented corrective actions.
- **Enabled business reporting:** Prepared structured, warehouse-ready datasets aligned with BI and reporting requirements to support analytics and decision-making.
- **Ensured controlled releases:** Managed code in Git and supported CI/CD workflows for controlled deployment of pipeline changes.

Analyst — Noida, India
HCL Tech — *Apr 2022 – Oct 2025*
Project: IT Operations Data Processing & Reporting
Tech Stack: Python, SQL, AWS S3, Excel, SharePoint, ServiceNow

- **Reduced manual data handling by 30%:** Automated recurring data preparation and reporting workflows using Python and SQL, improving processing consistency.
- **Built reporting-ready datasets:** Developed and maintained SQL datasets and Python scripts to extract, transform, validate, and consolidate IT operations and service-management data for networking, cybersecurity, and IT operations teams.
- **Protected data integrity:** Implemented validation and reconciliation checks to catch missing, duplicate, and inconsistent records across operational and compliance datasets, and resolved discrepancies through root-cause analysis and corrective action.

- **Improved processing efficiency:** Optimized SQL queries and Python scripts to streamline extraction, cleansing, and transformation for recurring reporting requirements.
- **Supported compliance and monitoring:** Prepared accurate datasets and resolved data issues within established SLAs for compliance reporting and operational monitoring.
- **Organized data storage:** Managed operational data files in AWS S3 and SharePoint for structured storage, retrieval, and downstream reporting.

Projects

Vendor Data Integration & ETL Pipeline
Python, SQL, Pandas, SQLAlchemy, Power BI, Git/GitHub

- **Built an ETL pipeline for 2.3M+ records:** Engineered a Python, Pandas, and SQLAlchemy pipeline that ingests 6 source datasets into a relational database, using chunked loading to handle the 2.3M-row sales file reliably.
- **Unified fragmented data:** Used SQL multi-table joins to combine purchase, sales, freight, vendor, and brand data into one analytics-ready dataset of **10,692** vendor-brand records across 18 fields, with zero null values.
- **Found and fixed a data integrity defect:** Traced invalid profit margin values in 178 zero-sales records back to the transformation logic and corrected the rules, so financial metrics stayed accurate and consistent downstream.
- **Validated and preserved data:** Applied cleansing, reconciliation, and validation checks, and flagged 698 dead-stock outliers (6.5% of records), filtering them out of margin averages instead of deleting them.
- **Delivered business-ready output:** Built reusable Python and SQL logic for vendor performance and inventory metrics, feeding a Power BI dashboard that tracks **$451.6M** in sales, a **28.7%** profit margin, and **$939.7K** in unsold capital.

E-Commerce Data Integration Pipeline
Python, SQL, PySpark, Azure Data Factory, Azure Databricks, Git

- **Unified e-commerce data:** Built ETL pipelines that consolidate product catalog, customer order, and sales transaction data into structured, analytics-ready datasets.
- **Ensured data consistency:** Applied cleansing, validation, and reconciliation (Python, SQL, PySpark) so related datasets agree across orders and sales.
- **Automated on Azure:** Orchestrated ingestion and transformation with Azure Data Factory and Databricks, with all code version-controlled in Git.

Returns & Refunds Data Pipeline
Python, SQL, PySpark, Azure Data Factory, Azure Databricks, Git

- **Consolidated returns data:** Integrated product returns, refund transactions, and return-status records into consistent operational datasets.
- **Improved tracking reliability:** Automated cleansing, transformation, and validation to flag inconsistent records before they affect returns reporting.
- **Enabled operational reporting:** Built repeatable workflows that deliver validated datasets for downstream reporting.

Payment Transaction Data Processing
Python, SQL

- **Consolidated payment data:** Developed Python and SQL logic to combine payment transactions, customer records, and refund data into structured datasets for financial reporting.
- **Protected financial accuracy:** Implemented transaction validation and reconciliation checks to catch inconsistencies and improve data reliability.
- **Prepared analysis-ready data:** Refined SQL transformations to produce consistent financial datasets for downstream analysis.

Education

Maulana Azad College of Engineering & Technology, Patna 2017 – 2021
Bachelor of Technology in Computer Science and Engineering

Certifications

AWS Cloud Practitioner Bootcamp (HCL Tech) Link
Power BI (PL-300) (HCL Tech) Link
Foundations: Data, Data, Everywhere (Coursera) Link
Ask Questions to Make Data-Driven Decisions (Coursera) Link
Machine Learning (Coursera) Link
Microsoft Excel (Coursera) Link

Extra Contributions

Volunteering: Received multiple appreciations from organizations for exceeding project expectations (2022–2025).